Introduction to Acids and Bases

Properties, models, and the language of acid–base chemistry

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Standards

Chemistry 2.6 — Acid–Base Chemistry: properties and defining models of acids and bases.

Objectives

1. What are the properties of acids and bases?
2. How do the Arrhenius, Brønsted-Lowry, and Lewis models define acids and bases?
3. What is a conjugate acid-base pair?

Expectations

1. Compare and contrast the three acid-base models
2. Identify monoprotic vs. polyprotic acids

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What Do You Already Know?

WHAT I KNOW	WHAT I WANT TO KNOW	EXPLORE THE SOURCE
▪ Lemons & vinegar taste sour ▪ Soap feels slippery ▪ Some cleaners are dangerous to mix	▪ Why do acids "burn"? ▪ Why does water conduct electricity a little on its own? ▪ What makes something a base if it has no OH^-?	► PhET: Acid-Base Solutions ► PhET: pH Scale ► Khan Academy: Acids & Bases *Tap a link to explore before class discussion.*

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Properties of Acids and Bases

ACIDS

- Taste sour
- Conduct electricity in solution
- Turn blue litmus paper red
- React with active metals (Mg, Zn) to release H_2 gas
- React with carbonates (e.g., limestone) to release CO_2
- Contain more H^+ ions than OH^- ions

BASES

- Taste bitter, feel slippery
- Conduct electricity in solution
- Turn red litmus paper blue
- React with some metals and carbonates too
- Contain more OH^- ions than H^+ ions
- Analogy: if an acid is a proton “donor,” a base is the friend willing to catch it

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Water's Balancing Act & the Arrhenius Model

Self-ionization of water

- Water produces equal numbers of H^+ and OH^- ions on its own.
- $H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

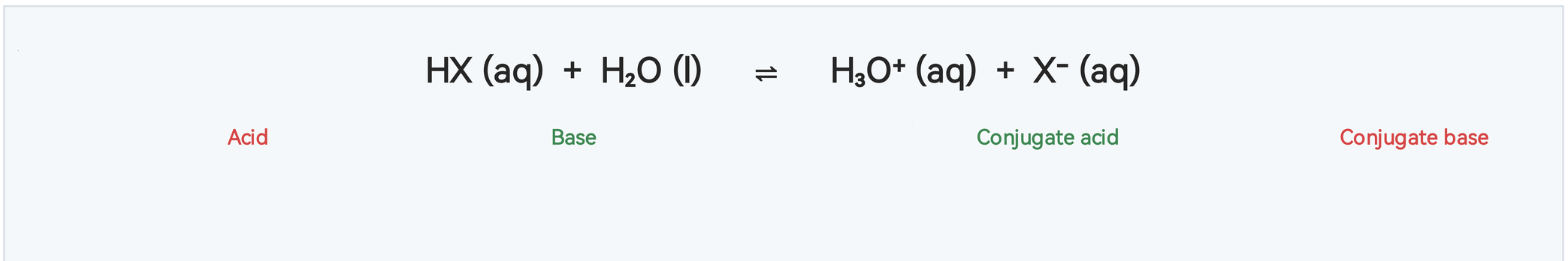
The Arrhenius Model

- An acid ionizes to form H^+: $HCl(g) \rightarrow H^+(aq) + Cl^-(aq)$
- A base dissociates to form OH^-: $NaOH(s) \rightarrow Na^+(aq) + OH^-(aq)$
- Limitation: NH_3 produces OH^- without containing a hydroxide group — Arrhenius can't explain that.

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The Brønsted–Lowry Model

- An acid is a hydrogen-ion donor; a base is a hydrogen-ion acceptor.
- More inclusive than Arrhenius — explains bases like NH_3 with no OH^- group.
- A conjugate base forms when an acid donates H^+. A conjugate acid forms when a base accepts H^+.
- Analogy: H^+ is a hot potato tossed from the acid to the base — whoever holds it becomes the conjugate acid.



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The Lewis Model — The Broadest View

- A Lewis acid is an electron-pair acceptor; a Lewis base is an electron-pair donor.
- Includes everything Brønsted-Lowry covers, plus reactions that never transfer H^+.
- **Model coverage: Arrhenius ⊂ Brønsted-Lowry ⊂ Lewis**
- Analogy: Lewis chemistry is a handshake — one substance offers an electron pair, the other accepts it.

$$H^+ + :F:^- \rightarrow H—F$$

Lewis acid: H^+

Lewis base: F^-

$$BF_3 + :NH_3 \rightarrow F_3B–NH_3$$

Water is amphoteric — it can act as either an acid or a base.

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Monoprotic & Polyprotic Acids

Acid	Formula	Conjugate Base	Formula
Hydrochloric acid	HCl	Chloride ion	Cl^-
Nitric acid	HNO_3	Nitrate ion	NO_3^-
Sulfuric acid	H_2SO_4	Hydrogen sulfate ion	HSO_4^-
Hydrofluoric acid	HF	Fluoride ion	F^-
Acetic acid	$HC_2H_3O_2$	Acetate ion	$C_2H_3O_2^-$
Phosphoric acid	H_3PO_4	Dihydrogen phosphate ion	$H_2PO_4^-$

Monoprotic acids (e.g., HCl) donate one H^+. Polyprotic acids (e.g., H_3PO_4) donate more than one, one at a time.

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Lesson Summary

Key Takeaways

1. Acids taste sour, conduct electricity, and turn litmus red; bases taste bitter, feel slippery, and turn litmus blue.

2. Water self-ionizes into equal H^+ and OH^-, the baseline every acid or base solution shifts away from.

3. Arrhenius: acids give H^+, bases give OH^-. Brønsted-Lowry: acids donate H^+, bases accept it. Lewis: acids accept an electron pair, bases donate one.

4. Monoprotic acids donate one H^+; polyprotic acids donate more than one, one at a time.

Test Yourself

1. Explain why many Lewis acids and bases are not classified as Arrhenius or Brønsted-Lowry acids and bases.

2. Compare and contrast the physical and chemical properties of acids and bases.

3. Explain how the concentrations of H^+ and OH^- determine whether a solution is acidic, basic, or neutral.

4. Identify the conjugate acid-base pairs: $HNO_2 + H_2O \rightleftharpoons NO_2^- + H_3O^+$

5. Challenge: The products of an acid-base reaction are H_3O^+ and SO_4^{2-}. Write the balanced equation and identify the conjugate pairs.

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